

SMART HOSPITAL CARE SYSTEM: IOT-Based Patient Monitoring and Alert System

TEAM MEMBERS:

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WEEKLY PROGRESS REPORT :

Week	Date	Work Done	Outcome
Week 1	25 Jul – 31 Jul	Problem identification, research and literature survey	Project problem and objectives finalized
Week 2	01 Aug – 07 Aug	Concept development and component selection	Basic system concept and components finalized
Week 3	08 Aug – 14 Aug	Hardware and software system design	System architecture and block diagram completed
Week 4	15 Aug – 21 Aug	Circuit, algorithm and design finalization	Hardware/software design completed
Week 5	22 Aug – 28 Aug	Started physical prototype; load cell, HX711 and STM32 setup	IV fluid monitoring started successfully
Week 6	29 Aug – 04 Sep	FSR integration, STM32–ESP32 communication and dashboard development	Bed monitoring and IoT dashboard integrated
Week 7	05 Sep – 10 Sep	System testing, debugging and validation	Integrated prototype tested and prepared for demonstration

WEEK 1 – PROBLEM IDENTIFICATION & RESEARCH

25 July – 31 July

- Identified the problem and project objectives.
- Studied existing hospital monitoring systems.
- Conducted literature survey.
- Defined the project scope and expected outcomes.
- Prepared initial project documentation.

WEEK 2 – CONCEPT DEVELOPMENT & SYSTEM DESIGN

01 August – 07 August

- Finalized the Smart Hospital Care System concept.
- Selected the required sensors and controllers.
- Planned IV fluid and patient bed monitoring.
- Prepared the initial system architecture.
- Defined hardware and software requirements.

WEEK 3 – HARDWARE & SOFTWARE DESIGN

08 August – 14 August

- Designed the overall system architecture.
- Prepared the functional block diagram and data flow.
- Planned STM32 and ESP32 communication.
- Designed the IV monitoring and bed monitoring sections.
- Prepared initial circuit and software design.

WEEK 4 – DESIGN FINALIZATION

15 August – 21 August

- Finalized the hardware and software design.
- Finalized circuit connections and component selection.

- Prepared algorithms for IV fluid monitoring and bed monitoring.
- Planned alert conditions and dashboard communication.
- Completed design and simulation preparation.

WEEK 5 – PROTOTYPE DEVELOPMENT

22 August – 28 August

- Started physical prototype development.
- Connected the load cell and HX711 with STM32.
- Tested load-cell readings using Docklight.
- Performed load-cell calibration.
- Started IV fluid weight and remaining percentage calculation.
- Implemented initial IV alert conditions.

WEEK 6 – PROTOTYPE INTEGRATION

29 August – 04 September

- Integrated FSR sensor for bed monitoring.
- Implemented patient occupancy and bed-exit detection.
- Completed IV monitoring calculations.
- Established STM32–ESP32 UART communication.
- Started Wi-Fi nurse dashboard development.
- Implemented patient admission and alert priority logic.

WEEK 7 – TESTING & VALIDATION

05 September – 10 September

- Tested IV fluid monitoring and alert conditions.
- Tested patient bed occupancy and bed-exit detection.
- Tested STM32–ESP32 communication.
- Tested nurse dashboard updates.
- Tested patient admission and alert priority.

- Tested acknowledge, resolved and alert history functions.
- Debugged and validated the integrated prototype.
- Prepared the system for final demonstration.